AI PRODUCTIVITY TRACKER

AI miniproject

Visvesvaraya Technological University, Belgavi.

PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

AI miniproject

SUBJECT CODE: BCS515B

Team	
Name	**USN**
Ashritha	4PM23CG006
Ayisha	4PM23CG007
Lekhana	4PM23CG022
Sahana	4PM23CG040
Shambhavi	4PM23CG045
Vijeta	4PM23CG057

PROJECT GOALS

- To develop an AI-based productivity tracker that helps users monitor and enhance daily performance.
- To analyze user activity patterns and provide insights, predictions, and anomaly alerts.
- To encourage better focus, goal completion, and efficiency through intelligent feedback.

PROBLEM & PROPOSED SOLUTION

Problem:

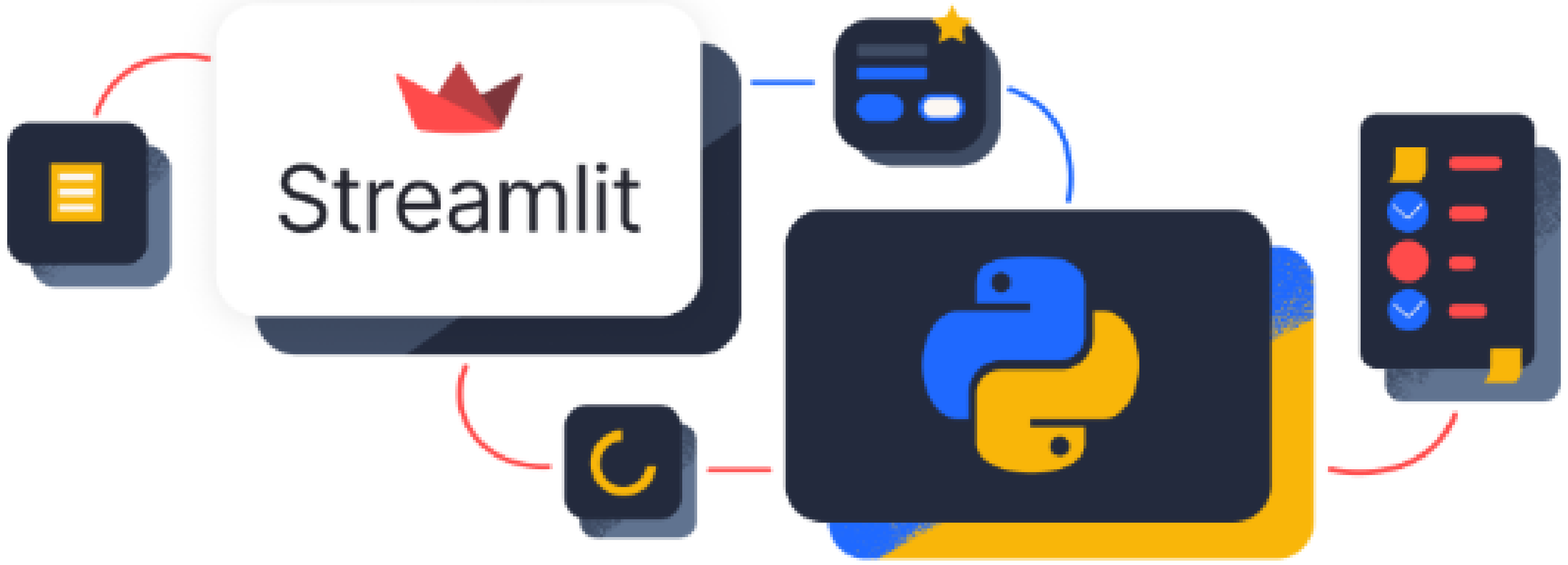
- People struggle to track and balance their daily tasks efficiently.
- Lack of insights into how time is spent on activities.
- Difficulty detecting unproductive habits or irregular routines.

Solution:

- An AI-powered productivity tracker that:
- Collects task data (to-do list, schedules, etc.)
- Applies Regression and Anomaly Detection algorithms
- Generates personalized reports, progress charts, and improvement tips.

AI TOOLS USED

1.Streamlit



2.ScikitLearn

AI MODELS USED

1. Regression Algorithm

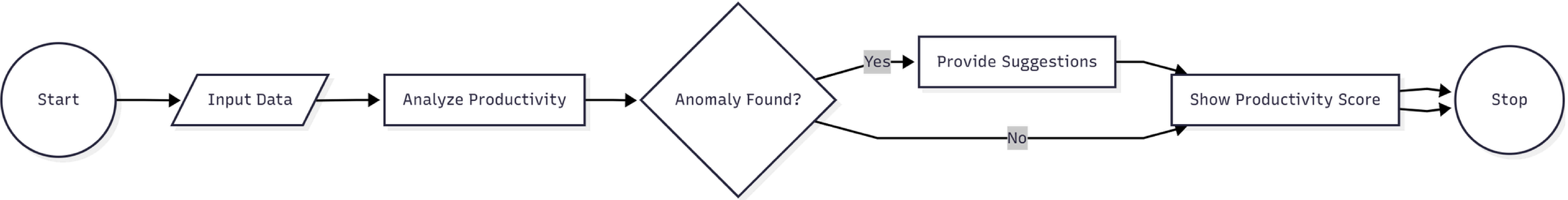
- Used to predict future productivity scores based on previous performance data.

- Algorithm: Linear Regression
  - y=mx+cy = m x + cy=mx+cx: number of completed tasks, time spent
  - y: productivity score
- Helps estimate how task completion rate affects productivity over time.

2.Anomaly Detection

- Detects unusual patterns in productivity data.
- Example: sudden drop in work hours, missed deadlines, or overworking.
- Algorithm Used: Isolation Forest
  - Works by isolating data points that differ significantly from normal behavior.
  - Flags them as anomalies for further review.
- Provides alerts and suggestions to help users maintain consistency.

FLOW CHART



Start
Input Data
Analyze Productivity
Anomaly Found?
Yes
Provide Suggestions
No
Show Productivity Score
Stop

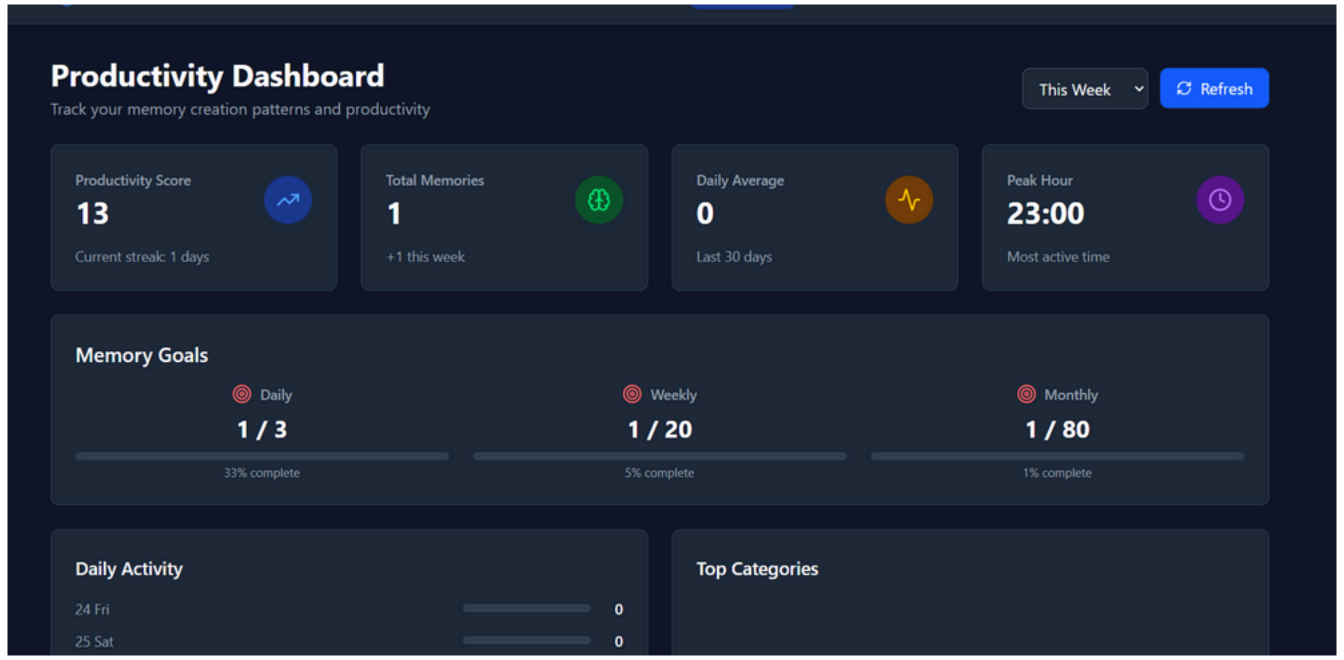
RESULT

Inputs:

1. To-Do List (tasks with priority)
2. Calendar Events (meetings, schedules)
3. Time Spent per Task (tracked automatically or manually)
4. Break Duration and Work Sessions
5. Productivity Rating (self-assessed daily)

Outputs:

1. Productivity Score Prediction
2. Daily/Weekly Performance Chart
3. Anomaly Alerts (low activity or excessive work)
4. Task Completion Statistics
5. AI Suggestions for Improved Efficiency



REAL-TIME APPLICATION

- **Workplace Productivity Monitoring**

Helps organizations like Google and Microsoft track employee focus time and identify distraction patterns for better efficiency.

- **Personal Time Management App**

Enables individuals to plan daily tasks and routines while AI predicts ideal working hours and suggests productivity improvements.

- **Student Study Planner**

Assists students in tracking study sessions, managing schedules, and maintaining focus through AI-based consistency analysis.

- **Remote Work & Freelancing Tool**

Supports freelancers in tracking project hours, analyzing productivity levels, and detecting burnout for a healthy workflow.

- **Health and Focus Assistant**

Integrates with wearables to monitor sleep, heart rate, and screen activity, offering real-time focus and wellness recommendations.

CONCLUSION

- The AI Productivity Tracker leverages Regression and Anomaly Detection to provide intelligent insights into personal efficiency.
- It enhances time management and promotes consistent improvement through adaptive feedback.

THANK YOU